See attached instructions.

https://gemini.google.com/app/3f7a358f71f43808

User prompt: See attached instructions.

Response: **Verdict: Do not approve revision 2; the design must return to its root.** Moving Study 2 to a self-financing account under Ville's inequality fixes the Student-t failure of revision 1, and pairing Study 1 on the same markets removes the uncalibrated point null. However, revision 2 introduces critical theoretical breaks in both studies: Study 2's mirror account tests a point-wise null ($\forall i, \mu_i \geq c_i$) whose negation is not portfolio loss, and its leverage map $\frac{1}{1-c}$ allows a genuinely profitable procedure to hit 20 on the mirror first and be falsely judged a net loser; the recommended half-Kelly stake rule is not a proper scoring rule for log wealth, hitting Brief 2's own wall by incentivizing 2x edge exaggeration while deploying roughly half the capital assumed in the power table; the pause rule (`cash + open cost ⩾ 20`) is invariant under purchases and permanently deadlocks if one settlement record is lost; the human's $2,500 reference stake on Kalshi's 7-day universe collapses admissible breadth below the 30-event/week abort wall; and Study 1's paired SPRT operating table (`arithmetic.py.txt`) was computed under the false assumption that the two arms are statistically independent on the same market.

Ranked Findings (Part 1 of 2)

1. Critical — Study 2's mirror account tests the wrong null, subsidizes fees, and falsely convicts profitable procedures

- **Where:** `architecture.md` §5.2: *"The mirror. A second account, computed from the same record, buys for each purchase the complementary claim, which pays `1 - x`, at `1 - c`, staking the fraction of its own cash that the account staked of its cash. Under the hypothesis that every purchase had non-negative expected net return it is a non-negative supermartingale. If it reaches 20, that hypothesis is rejected at 5% and the procedure is said to lose net of costs."* Also §4, §8, and `paper-changes.md` A7.

- **Construction:**

  1. **Logical mismatch between null and claim:** The main account's null is $H_0^+ : \forall i,\ E[x_i \mid \mathcal{F}_{t_i}] \leq c_i$, whose negation is $\exists i,\ E[x_i \mid \mathcal{F}_{t_i}] > c_i$. The mirror account's null is $H_0^- : \forall i,\ E[x_i \mid \mathcal{F}_{t_i}] \geq c_i$. The negation of H_0^- is **not** that *"the procedure's purchases lose net of costs"*; it is merely $\exists i,\ E[x_i \mid \mathcal{F}_{t_i}] < c_i$ (at least one purchase had negative expected net return). Any real procedure with heterogeneous bets violates both H_0^+ and H_0^- simultaneously.

  2. **Leverage distortion $\frac{1}{1-c}$ vs. $\frac{1}{c}$:** When the main account stakes fraction f of cash at cost c, its payout multiplier per unit is $\frac{1}{c}$. The mirror stakes the same fraction f of

mirror cash at price $1-c$, giving payout multiplier $\frac{1}{1-c}$. On a favorite at $c=0.95$ with $f=0.05$, a loss ($x=0$) costs the main account 5% of cash, but pays the mirror account $0.05\times\frac{0.95}{0.05}=+95\%$ of its entire cash (nearly doubling the mirror in one bet). Conversely, on a cheap contract at $c=0.20$, a win pays the main account $4f$ but costs the mirror only f, while a loss pays the mirror only $\frac{0.20}{0.80}f=0.25f$.

3. **Profitable run that fails via the mirror:** Suppose each week the procedure makes 30 purchases at all-in cost $c=0.20$ where true win probability is $q=0.22$ (a $+10\%$ net edge per dollar staked), staking half-Kelly $f_1=\frac{0.22-0.20}{2(0.80)}=0.0125$ or a moderate $f_1=0.03125$ of cash each, and 5 purchases on favorites at $c=0.95$ where the agent reports $p=0.96$ ($f_2=0.05$) but true probability is $q=0.88$ (a -7.37% net loss per dollar on those 5 bets).

   - **Main account:** Weekly expected net return is $30(0.03125)(+0.10)+5(0.05)(-0.0737)=+0.09375-0.01842=+\mathbf{7.53\%}$ of cash per week. Weekly expected log growth is $30[0.22\ln(1.125)+0.78\ln(0.96875)]+5[0.88\ln(1.00263)+0.12\ln(0.95)]=+0.0393-0.0192=+\mathbf{0.0201}$ per week (reaches 20 in ~ 149 weeks).

   - **Mirror account:** On the 30 bets at $1-c=0.80$ ($1-q=0.78$), weekly log growth is $30[0.78\ln(1.00781)+0.22\ln(0.96875)]=-0.0273$. On the 5 bets at $1-c=0.05$ ($1-q=0.12$), weekly log growth is $5[0.12\ln(1.95)+0.88\ln(0.95)]=+\mathbf{0.1750}$. Net mirror log growth is $+0.1750-0.0273=+\mathbf{0.1477}$ per week—reaching 20 in 20.3 **weeks** (with probability $>99\%$ by week 52). Under §8 (*"ends with the claim when cash reaches 20, and as a loss when the mirror does"*), the study terminates around week 20 and falsely declares a $+7.53\%$/week profitable procedure a net loser.

4. **Fee and spread subsidy:** Because $c=a+\text{fee}$ is the all-in ask cost, pricing the mirror at $1-c=(1-a)-\text{fee}$ instead of the executable complement ask $(1-b)+\text{fee}$ subsidizes the mirror by $(a-b)+2\,\text{fee}$ per share.

- **Outcome:** False negative ("loses net of costs") judgments against genuinely profitable procedures, or simultaneous crossing of 20 by both accounts (for which §8 defines no tie-break).

- **Severity:** Critical.

2. Critical — Study 2's half-Kelly stake rule is improper (hitting Brief 2's wall), under-deploys capital relative to the power table, and deadlocks under the pause rule

- **Where:** `architecture.md` §5.3: *"Recommended start: half of Kelly's fraction from the agent's own registered probability `p`, `(p - c) / (2(1 - c))` of cash, capped at 5% of cash, and no purchase unless `p > c`. Log wealth under Kelly's rule is a proper score, so the agent's honest probability is its best report."* `briefs/2-thesis-track.md` Q4: *"If the rule gives the agent a reason to misreport its probability, that is a wall."* `architecture.md` §5.3: *"The pause. When cash plus the cost of open positions reaches 20, the account stops buying until they settle."* `architecture.md` §5.7 property 2.

- **Construction:**

  1. **Half-Kelly is not a proper scoring rule for log wealth:** For true probability $q > c$, expected single-bet log growth $G(f) = q\ln\left(1 + f\frac{1-c}{c}\right) + (1-q)\ln(1-f)$ is uniquely maximized at the **full-Kelly** fraction $f^*(q) = \frac{q-c}{1-c}$. Under the half-Kelly rule $f(p) = \frac{p-c}{2(1-c)}$, an agent with true belief q achieves $f(p) = f^*(q)$ if and only if it **misreports**

  $$p^* = c + 2(q-c) = 2q - c \neq q.$$

  For $c = 0.50$ and true probability $q = 0.52$, reporting honestly ($p = 0.52 \implies f = 0.02$) yields expected log growth 0.00060, whereas exaggerating the edge two-fold ($p = 0.54 \implies f = 0.04$) yields 0.00080 (33% faster log growth). By Brief 2 Q4, this is an immediate design wall.

  2. **Disconnect between §5.3's stake rule and §5.4's power table:** Section 5.4 (`arithmetic.py.txt`, `account()`) assumes `deployed = 1.0` (100% of cash staked in equal $1/m$ parts every week). Under §5.3's actual rule $f = \min\left(0.05, \frac{p-c}{2(1-c)}\right)$ applied to remaining cash at each registration, an honest agent with the table's stated edges never reaches the 5% cap (which requires $p - c \geq 0.10(1-c)$):

     - In the table's row `60 at 0.20 | 0.10 (2 points)` ($p = 0.22, c = 0.20$), the rule stakes $f = \frac{0.02}{2(0.80)} = \mathbf{1.25\%}$ of remaining cash per purchase. Across 60 sequential purchases, total deployed cash is $1 - 0.9875^{60} = \mathbf{53.0\%}$ (not 100%), cutting weekly log growth from 0.064 to 0.043, raising expected weeks from **47 to 70**, and cutting 52-week power from **0.66 to 0.24**. With 30 purchases, it deploys only 31.5% of cash, failing §5.4's 50% shakedown wall.

     - In row `30 at 0.50 | 0.08` ($p = 0.54, c = 0.50$), $f = 0.04$, deploying $1 - 0.96^{30} = 70.6\%$ of cash; with 2 points of edge ($p = 0.52$), $f = 0.02$, deploying only 45.5% of cash (again failing the 50% shakedown wall).

3. **The pause rule is purchase-invariant and deadlocks on a single lost settlement record:** Let C_t be cash and $S_t = \sum_{i \in \text{open}(t)} s_i$ be the cost of open positions. Any purchase of stake s maps $(C_t, S_t) \mapsto (C_t - s, S_t + s)$, leaving $C_t + S_t$ **strictly invariant**. Thus $C_t + S_t$ never reaches 20 during purchases—even when open positions have massive positive expected value—and only changes when a position settles. Conversely, suppose a settled win pushes $C_t + S_t$ to 20.1 ($C_t = 19.5, S_t = 0.6$) while one open position has a lost or indefinitely disputed settlement record. Because the pause stops buying *"until they settle"*, the account is **permanently frozen at** $C_t = 19.5 < 20$ and can never reject the null, regardless of edge.

4. **Undefined domain when $c \geq 1$:** Walking thin asks near 0.99 plus fees makes all-in cost $c \geq 1$, violating `c in (0, 1)` and causing division by zero or negative stakes in `(p - c)/(2(1 - c))` and `1 - c`.

- **Outcome:** Breaks Brief 2's proper-scoring requirement, overstates 52-week power by $2\times$ $-3\times$, triggers false shakedown aborts, and allows a single missing settlement record to permanently deadlock a winning account below 20.

- **Severity:** Critical.

3. Critical — The human's $2,500 reference stake on Kalshi's 7-day universe collapses breadth below the abort threshold and turns gross edges negative

- **Where:** Prompt (Human's answers to §12): *"Study 2 runs, on Kalshi. The reference stake is $2,500 per purchase, costed against the recorded book… The agent does not fetch result pages: the certified search API must return page content itself."* `architecture.md` §5.3–5.4: *"The reference stake fills from the recorded asks. Otherwise the registration is refused. The contract's scheduled end is within a horizon cap. Seven days is the starting point. One open position per event… fewer than 30 distinct admitted events settling a week"*.

- **Construction:**

1. **Contract depth required by $2,500:** Buying $2,500 at all-in cost c requires walking $\frac{2{,}500}{c}$ contracts on the ask book: **5,000 contracts** at $c = 0.50$, **12,500 contracts** at $c = 0.20$, and **25,000 contracts** at $c = 0.10$.

2. **Collapse of Kalshi's 7-day universe:** Kalshi's short-horizon listings are dominated by multi-strike ladders (daily/weekly equity indices, BTC/ETH, Fed rate tiers, city temperatures, economic prints) and sports. Collapsing ladders to *"one open position per event, by the venue's own event grouping"* reduces 20–50 strikes on one underlying date to **1 event**. Outside sports and HFT-priced index/crypto ladders, reading-amenable catalyst events settling within 7 days that carry $\geq \$2,500$ of ask depth near top-of-book number **fewer than 5 to 15 per week**—failing §5.4's pre-freeze wall (*"fewer than 30 distinct admitted events settling a week"*).

3. **No slippage cap in §5.3:** Because §5.3 requires only that the $2,500 stake *"fills from the recorded asks"* with no spread/slippage limit, a thin market with 200 contracts at 0.48 and stub asks at 0.85–0.99 "fills" $2,500 at an average cost $c \approx 0.80$, turning a true probability $q = 0.54$ into an un-registerable ($p < c$) or deeply negative-return trade.

4. **Turnover and power impact:** If only $m = 8$ non-sports events/week pass the $2,500 depth and $p > c$ filters at average stake $f = 0.03$ (deploying $1 - 0.97^8 = 21.6\% < 50\%$ of cash, failing the shakedown wall), and a 4-point gross probability edge ($q - a_{\text{top}} = 0.04$ at 0.50) pays 1.75 cents in Kalshi taker fees plus 1.50 cents of $2,500 book-walk slippage, net edge is 0.75 cents ($e = 0.015$). Weekly log growth is $8[0.03(0.015) - \frac{1}{2}(0.03)^2] = \mathbf{0.0000}$ (**infinite expected time to 20**; even at $e = 0.04$, expected time is **500 weeks / 9.6 years**, 52-week power < 0.001).

- **Outcome:** Study 2 either aborts before freezing under §5.4's census/shakedown walls or runs with near-zero power because $2,500 book walks consume the edge on thin 7-day events.

- **Severity:** Critical.

4. Critical — Study 1's paired SPRT table (`arithmetic.py.txt`) falsely assumes the Agent and G100 are independent on the same market

- **Where:** `architecture.md` §6: *"Test: Wald's sequential test of `θ ≤ 1/2` against `θ = 0.8` (the agent at 0.5 against G100 at 0.2), nominal errors 5%, ending undecided at 60 markets. By exact enumeration: [table]"*; `arithmetic.py.txt` lines 54–56: `cases.append((f"agent {agent}, G100 {searcher}", agent * (1 - searcher), searcher * (1 - agent)))`.

- **Construction:**

1. For paired binary outcomes (A_m, B_m) on the **same** market m with marginals $p_A = P(A_m = 1)$ and $p_B = P(B_m = 1)$, let $p_{11} = P(A_m = 1, B_m = 1)$. Then discordant probabilities are $p_{10} = p_A - p_{11}$ and $p_{01} = p_B - p_{11}$, and

$$\theta = \frac{p_A - p_{11}}{p_A + p_B - 2p_{11}}, \qquad p_{\text{disc}} = p_A + p_B - 2p_{11}.$$

   `arithmetic.py.txt` hardcodes $p_{11} = p_A p_B$ (**statistical independence on the same market**), contradicting the entire reason for pairing.

2. **Failure across the allowed `[0.1, 0.3]` calibration band:** Section 6 accepts any calibration strength where G100's rate is in `[0.1, 0.3]`. Even under

`arithmetic.py.txt`'s own independence formula, if G100's rate is $p_B = 0.30$ and the Agent hits the target $p_A = 0.50$, then $\theta = \frac{0.35}{0.35+0.15} = \mathbf{0.70} < \mathbf{0.80}$: upper-decision power drops from **0.888 to 0.685**, and the false lower-decision rate jumps from **0.033 to 0.142** ($2.8\times$ the nominal 5% error). With negative association ($p_{11} = 0.05$), $\theta = 0.643$, power drops to **0.486**, and the lower decision occurs **30.8%** of the time.

3. **False target-advantage claim under positive correlation:** Because easy market seeds make the plant easier for both arms ($p_{11} > p_A p_B$), suppose $p_A = 0.25$ and $p_B = 0.20$ with $p_{11} = 0.1875$ ($p_{10} = 0.0625, p_{01} = 0.0125$). Then $\theta = \frac{0.0625}{0.075} = \mathbf{0.833} > \mathbf{0.80}$: an agent that beats G100 by only 5 percentage points (0.25 vs. 0.20) exceeds the $\theta = 0.80$ alternative designed for $(0.50, 0.20)$.

- **Outcome:** Invalidates Section 6's power and Type-II error table and decouples $\theta = 0.80$ from the stated $(0.50, 0.20)$ capability target.
- **Severity:** Critical.

CONTINUE FOR PART 2

User prompt: continue

Response: Ranked Findings (Part 2 of 2)

5. High — Study 2's validity proof and Property 2 fail under proportional staking with delayed payouts, and `arithmetic.py.txt` simulates simultaneous rather than sequential stakes

- **Where:** `architecture.md` §5.1: *"Under the null, cash plus the conditional expected payout of the open positions is a non-negative supermartingale that starts at 1: each purchase replaces `s` of cash with a claim worth at most `s`, and between purchases a conditional expectation is a martingale."* `architecture.md` §5.7 Property 1 & 2: *"Both accounts are pure functions of the recorded bytes and the manifest… With the purchases and their stake fractions held fixed, cash at every time is non-decreasing in each recorded payout: a lost settlement record lowers it and a late one delays it… So no failure after a registration can raise the chance of a false rejection above 5%."* `arithmetic.py.txt` lines 78–80 (`account`).
- **Construction:**

1. **Delayed settlement records inflate cash under proportional staking (violating the premise of Property 2):** Property 2 holds purchases and stake fractions f_t fixed,

claiming that a late settlement record only delays cash and never increases it. That is false when stakes are proportional to current cash ($s_t = f_t C_t$). Suppose at $t = 1$ the account has cash $C_1 = 1.80$ and an open position settles with payout $P_1 = 9.50$. From $t = 2$ to $t = 21$, the procedure makes 20 losing purchases, each staking $f = 0.05$ of current cash, followed at $t = 22$ by a winning purchase that doubles cash ($\times 2$):

- **On-time settlement record at $t = 1$:** Cash becomes $C_1 = 1.80 + 9.50 = 11.30$. The 20 losing bets at 5% of current cash reduce cash to $11.30 \times (0.95)^{20} = 4.051$ at $t = 21$. Doubling at $t = 22$ gives terminal cash $8.10 < 20$ (does not reject).
- **Late settlement record delayed until $t = 22$:** At $t = 1$, cash remains 1.80 while $P_1 = 9.50$ sits unrecorded in the settled position, shielded from proportional drawdowns. The 20 losing bets reduce 1.80 to $1.80 \times (0.95)^{20} = 0.645$. When the polling daemon records $P_1 = 9.50$ at $t = 22$ ($C = 10.145$) before the $\times 2$ win, terminal cash reaches $20.29 \geq 20$ **(rejects the null)**. Thus, cash is **not** non-decreasing in timely payout recording: operational polling lag during a losing streak shields realized payouts from proportional staking losses and manufactures a crossing of 20 that on-time recording would have prevented.

2. **The null is stated only at registration, not across operational polling times:** Section 5.1 states the null as $E[x_i \mid \mathcal{F}_{\tau_i}] \leq c_i$ at registration τ_i. For $W_t = C_t + \sum_{i \in \text{open}(t)} \frac{s_i}{c_i} E[x_i \mid \mathcal{F}_t]$ to be a supermartingale when the operational layer chooses when to poll settlements or retry failed fetches, the filtration $\mathcal{F}_t$ must make polling delays stopping times independent of unrecorded outcomes x_i. If an operational failure (e.g., HTTP 429 or schema change on disputed/split markets) correlates with outcome state x_i, $\tilde{M}_t^{(i)} = E[x_i \mid \mathcal{F}_t]$ is not a martingale on the recorded byte stream.

3. **Simultaneous vs. sequential staking in `arithmetic.py.txt`:** In `account()`, `factor = 1 - deployed + deployed * wins / (bets * cost)` models $m = 60$ bets placed **simultaneously** at the start of the week, each receiving $\frac{\text{deployed}}{m}$ of initial weekly cash. Under §5.1 and §5.3, however, purchases are registered sequentially from remaining cash: staking f across m sequential purchases before settlement deploys $1 - (1 - f)^m < 1$ of cash (`small_numbers()` uses $1 - 0.95^{30} = 0.785$), never `deployed = 1.0`.

- **Outcome:** Operational polling delays can convert a non-rejecting path (8.10) into a rejecting path (20.29), and the power script simulates a simultaneous batch portfolio that the sequential account cannot execute.

- **Severity:** High.

6. High — Study 1's "pick" and alignment measure are corrupted by the frozen kernel's bankruptcy resurrection, scale-invariant noise, and a masked leak check

- **Where:** `architecture.md` §6: *"A session's pick is its evaluated program with the highest search-period score, the quantity G100 maximizes, read from the ledger the same way for both arms. The pick carries the plant when the alignment of its holdings with the planted signal is at least half the locked reference program's on the same market… Recommended start: the cosine between dollar holdings and the signal over all coins and steps"*; *"Model-free checks… 2. On zero-plant markets… G100's rate is at most a quarter of its rate at the chosen strength… 3. G100's picks from zero-plant markets, re-run on fresh zero-plant markets, average a net score at or below zero. An excess is a leak in the evaluator."* Also `argon-paper-findings.md` Finding 2.

- **Construction:**

  1. **Bankruptcy resurrection (Argon Finding 2c) corrupts Study 1's pick and holdings:** In `argon-paper-findings.md` Finding 2, Fable claims against GPT that *"study 1 judges holdings and reads no return, so absorption matters there only for when a holding ends."* That is false. Study 1 runs the **frozen pilot kernel** (§7) and selects each session's **pick** by *"the highest search-period score"*—which is a **return-based score** computed by the defective pilot kernel. A program that goes bankrupt mid-window and resurrects (Finding 2c) changes both its return score (hijacking which program becomes the session's pick) and its post-bankruptcy `Fill` events in the trace.

  2. **Cosine alignment ignores catastrophic losses and unplanted-coin churn:** Because $\cos(H, S) = \frac{\langle H, S \rangle}{\|H\|_2 \|S\|_2}$ is scale-invariant and reads no returns, a strategy that holds $\$10$ aligned with the signal on the planted coin for 1 step, gets liquidated or pays $50\times$ the plant's value in taker fees, and holds $\$0$ thereafter achieves the same cosine alignment as a profitable strategy holding $\$10,000$. Furthermore, if the reference program gets stopped out early on a volatile market seed so that $\cos(H_{\text{ref}}, S) = 0.04$, any noisy pick with $\cos(H, S) \geq 0.02$ passes the half-reference threshold.

  3. **Model-free Check 3 masks evaluator leaks smaller than transaction costs:** On fresh zero-plant markets, any trading strategy pays strictly positive fees and slippage, so a leak-free evaluator gives G100's picks a **negative** expected net score $-\mu_{\text{cost}} < 0$ (typically -1.5% to -3.0% annualized Sharpe-equivalent). Requiring only that average net score be *"at or below zero"* allows an evaluator lookahead leak of $+\mu_{\text{leak}} \leq \mu_{\text{cost}}$ (e.g., Sharpe $+1.5$) to pass Check 3 with probability $> 95\%$.

4. **Model-free Check 2 lacks sample size and fails to bound discordant bias:** Neither §6 nor Brief 1 specifies how many zero-plant markets Check 2 runs or with what confidence bound, and a false-positive alignment rate of $\frac{1}{4}(0.20) = 0.05$ on G100 does not bound the agent's false-positive alignment rate if the agent proposes simpler, lower-dimensional programs than G100's genetic trees.

- **Outcome:** Resurrected bankrupt programs or fee-burning strategies can be scored as "carrying the plant," while lookahead leaks up to the strategy's fee drag pass Check 3 undetected.
- **Severity:** High.

7. High — The confirmation wall (4.98 vs. 26) and Table 1 Row G conflate 80%-power thresholds with $\sim 50\%$-power expected log-wealth times

- **Where:** `architecture.md` §2.1–2.3: *"At T = 0.25 years a test that trusts the Gaussian approximation has 80% power only at Sharpe 4.98, and the gate's exact test needs about 26."* `architecture.md` §3 Table Row G: *"net edge per dollar staked 0.04; 0.08; 0.12; 0.16 | exact: 22; 10; 6.5; 5 (expected weeks to 20: 95; 43; 28; 21)"*. `arithmetic.py.txt` (`yardstick`, `capped_bet`, `account`).
- **Construction:**
  1. **Apples-to-oranges power comparison (4.98 at 80% power vs. 26 at $\sim 50\%$ power):**
     - In `yardstick()`, $4.98 = \frac{z_{0.95}+z_{0.80}}{91/365} = \frac{2.4865}{0.2493}$ is the Sharpe for **80% power**.
     - In `capped_bet()`, 26.03 solves $n\left(\lambda\mu - \frac{1}{2}\lambda^2(\sigma^2+\mu^2)\right) = \ln 20 = 2.9957$ for $n = 546, \lambda = 0.5, \sigma = 0.02$. That equates **expected log wealth** $E[\ln K_n]$ to $\ln 20$, which corresponds to roughly **50% power**, not 80% power. Because $\mathrm{SD}(\ln K_{546}) \approx \quad 546\lambda\sigma = 0.2337$, achieving **80% power** ($\ln 20 + 0.8416 \times 0.2337 = 3.1924$) requires Sharpe **27.7** (and over 2 years requires **4.0**, not **3.4**).
     - Even 27.7 assumes a constant oracle bet $\lambda = 0.5$ from step 1 and a quadratic Taylor approximation $\lambda\mu - \frac{1}{2}\lambda^2(\sigma^2+\mu^2)$ that ignores negative skewness $\gamma_3 < 0$ and heavy kurtosis $\gamma_4 > 3$; under the gate's actual plug-in bet (Argon Finding 9), which starts at 0 and jumps, required Sharpe for 80% power exceeds **30**.
  2. **Section 3 Table Row G reports expected hitting times ($\sim 50\%$ power) in an 80%-power column:** Rows A–E of the Section 3 table report months to **80% power** (`6.18 / (d^2 * n)`). Row G reports `22; 10; 6.5; 5` months by dividing

`expected weeks to 20: 95; 43; 28; 21` by 4.33. As §5.4's own simulation table proves, at edge `0.04`, 104 weeks (**24 months**) achieves only **0.65 power** (not 0.80, which requires **28 months** at 100% weekly cash turnover and **56 months** at 50% turnover).

- **Outcome:** Understates the gate's true 80%-power Sharpe threshold and makes Study 2 appear faster in Section 3's comparison table by reporting $\sim 50\%$-power expected hitting times alongside 80%-power yardstick months.
- **Severity:** High.

8. Medium — Fable's dispositions of Argon's paper findings 4, 7, 10, and 11 in `argon-paper-findings.md` rely on flawed math

- **Where:** `argon-paper-findings.md` Findings 4, 7, 10, and 11; `arithmetic.py.txt` lines 100–125 (`argon`).
- **Construction:**

1. **Finding 11 (Stationary bootstrap block length — wrongly rejected by Fable):** Fable writes: *"Does not hold as stated… Under the null the returns of a predictable position are martingale differences: funding persists, the return's mean does not."* This is mathematically false for the gate's search screens (White's Reality Check, Hansen's SPA, Romano–Wolf StepM). Those screens test the **unconditional** null $E[r_{k,t}] \le 0$ by bootstrap-resampling centered returns $r_{k,t} - \bar{r}_k$. In both N2 and real perps, a strategy holding a constant or slow-moving position $\text{pos}_{t-1} = +1$ receives persistent 4-hour funding f_t (with AR(1) coefficient $\phi = 1 - 1/42 = 0.9762$). Because price innovations in N2 are independent of f_t (outside the plant), the strategy's total return is $r_t = \epsilon_t - f_t - \text{cost}_t$, whose **conditional mean** $E[r_t \mid \mathcal{F}_{t-1}] = -f_t - \text{cost}_t$ **persists with** $\phi = 0.9762$! Therefore, r_t is **not** a martingale difference under the null: its autocovariance at lag h is $\text{Cov}(r_t, r_{t+h}) = \text{Var}(f_t)\phi^h > 0$, inflating long-run variance by $\frac{1+\phi}{1-\phi} = 83\times$ on the funding component (and $8.5\times$ when funding is 10% of return variance), of which block length $L = 6$ captures only $\frac{1+\phi(5/6)}{1-\phi(5/6)} = 9.7$ out of 83.
2. **Finding 10 (v3 locked suite tolerances — Fable sneaks in a 99-way Bonferroni correction):** Fable writes: *"By exact Clopper–Pearson bounds, Argon's two examples (160 and 248 rejections) fail, narrowly."* Look at `arithmetic.py.txt` line 117: `beta.ppf(1 - 0.05 / 99, x + 1, 20_000 - x) ⩽ tolerance`. Fable divided $\alpha = 0.05$ by **99** ($\alpha = 0.000505$, a 99.9495% upper bound) to get max passing counts of `154` and `244`. Under the standard 95% Clopper–Pearson bound (`beta.ppf(0.95, x + 1, 20_000 - x)`), $x = 160$ gives **0.00915** ≤ 0.01 (passes up to **178** rejections), and $x = 248$ gives **0.01379** ≤ 0.015 (passes up to

269 rejections). Even at 99% confidence, 160 gives $0.00965 \leq 0.01$ and 248 gives $0.01440 \leq 0.015$.

3. **Finding 7 (Deflated Sharpe Ratio ceiling — Fable checks only $\hat{S} \to \infty$ at zero skewness):** Fable claims the DSR ceiling falls below 0.95 *"only above a kurtosis of about `1 + 1.48(T - 1)`."* That solves $\lim_{\hat{S}\to\infty} \mathrm{DSR}(\hat{S}) < 0.95$ at $\hat{\gamma}_3 = 0$. At any realistic finite per-step Sharpe—e.g., annualized $\hat{S}_{\text{ann}} = 3.5$ ($\hat{S} = 0.0749$ per 4h step) against threshold $S_{0,\text{ann}} = 2.0$ ($S_0 = 0.0428$) over $T = 546$ steps—the numerator is $(\hat{S} - S_0) \quad 545 = 0.749 < 1.64485$, so $\mathrm{DSR} < 0.95$ **at every kurtosis** $\hat{\gamma}_4 \geq 1$, and even for $\hat{S}_{\text{ann}} = 6.0$ ($\hat{S} = 0.128$) with negative skewness $\hat{\gamma}_3 = -4$, DSR falls below 0.95 at $\hat{\gamma}_4 = 18 \ll 546$.

4. **Finding 4 (Stage 4 campaign recount):** Fable counts v2's two zero-evaluation crashes as valid gate tests. A campaign that crashes with **zero evaluations** has an identically zero chance of promoting a strategy (`0/0`); counting zero-evaluation crashes in a gate false-promotion bound dilutes the per-campaign promotion rate with non-runs.

- **Outcome:** Leaves the stationary bootstrap's block-length underestimation unacknowledged in the defect table and puts false 99-way Bonferroni numbers (`1,138, 244 and 154`) into `paper-changes.md` B.

- **Severity:** Medium.

9. Medium — `paper-changes.md` introduces four new inconsistencies and mathematical errors into the whitepaper

- **Where:** `paper-changes.md` edits A2.3, A5.5, A6, A7.2, A7.3, and B (Finding 10).

- **Construction:**

1. **Contradictory log-growth formulas between `architecture.md` §2.2 and `paper-changes.md` A7.3:** For m simultaneous even-price bets of net edge e, portfolio return $\bar{r} = \frac{1}{m}\sum_{i=1}^{m} r_i$ has mean e and variance $\frac{1-e^2}{m}$. At full turnover ($\lambda = 1$), the second-order Taylor approximation to expected log growth is $e - \frac{1-e^2}{2m} - \frac{e^2}{2}$. `architecture.md` §2.2 writes `e - 1/(2m)` (dropping $-e^2/2$, which gives 0.0717 at $e = 0.08, m = 60$), while `paper-changes.md` A7.3 writes `e - 1/(2m) - e²/2` (0.0685), and neither matches exact binomial log growth (`0.0689`) or §5.3's sequential half-Kelly rule.

2. **Edit B (Finding 10, S4 line 645):** Inserts `"Up to 1,138, 244 and 154 rejections in 20,000 pass"` into S4 without disclosing that those thresholds

require a $\frac{0.05}{99}$ Bonferroni correction across 99 tests rather than the single-cell Clopper–Pearson bound (1, 182, 269, and 178).

3. **Edit A5.5 (Table `tab:rates`):** Places Row G's expected weeks to 20 ($\sim 50\%$ power under 100% weekly turnover) into a table whose caption and other rows represent $6.18/(d^2 n)$ (**80% power**).

4. **Edit A7.2:** Formalizes the mirror account proof in Appendix T4 while retaining the false claim in A5.7 that the mirror reaching 20 proves the procedure loses net of costs (Finding 1).

- **Outcome:** Introduces new mathematical errors into `docs/whitepaper.tex` if applied as written.

- **Severity:** Medium.

10. Medium — The human's §12 search-API constraint and undecided Polymarket/perp records create immediate compliance and survivorship blockers

- **Where:** Prompt (Human's answers to §12): *"The agent does not fetch result pages: the certified search API must return page content itself… The perp instance and the Polymarket recordings are undecided."* `architecture.md` §5.6 item 1–3: *"1. an official API whose terms permit automated queries and the use of results as input to a language model; 2. no clause against storing responses… 3. page content returned by the API itself, since the daemon never fetches a result's URL from an uncertified host."*

- **Construction:**

1. **Third-party copyright pass-through on full page content:** Requiring the search API to return full page content (rather than snippets/links) shifts the terms audit from the search vendor to the underlying publishers. Search APIs that return full scraped page bodies or cached text (e.g., Exa, Tavily, Brave, Perplexity/Serp wrappers) uniformly disclaim copyright licenses over third-party publisher text in their ToS, passing liability for permanent hash-addressed storage (`item 2`) and LLM ingestion (`item 1`) to the operator. Under the project's strict compliance rule, no search API can grant a sublicense to store and feed full third-party news articles that it does not own.

2. **Undecided Polymarket recordings:** Because Polymarket's August 11, 2026 terms forbid automated scraping and data extraction (GPT Finding 9), retaining and reading previously scraped Polymarket files for any future calibration or paper diagnostic violates the compliance policy unless explicitly purged or isolated under an operator legal decision.

3. **Undecided perp instance:** Keeping the perp recorders or daily sessions running while parked consumes part of the 600 weight/min Hyperliquid budget and personal

account allowance for a track that §2.4 proves requires $618–2,473$ independent pairs ($3–10$ years).

- **Outcome:** Blocks Study 2's freeze under §5.6 item 2–3 unless the search API is restricted to licensed/public-domain sources or short fair-use metadata snippets, and leaves uncertified Polymarket data on disk.
- **Severity:** Medium.

Summary Table of Ranked Findings

Rank	Severity	Target	Core Theoretical / Operational Break
1	**Critical**	Study 2 mirror account (§5.2, §8, A7)	Negating $\forall i, \mu_i \geq c_i$ is $\exists i, \mu_i < c_i$, not portfolio loss; $\frac{1}{1-c}$ leverage and fee subsidy let a $+7.53\%$/wk profitable procedure hit 20 on the mirror in 20.3 weeks and be falsely declared a loser
2	**Critical**	Study 2 stake & pause rules (§5.3, §5.4, Brief 2 Q4)	Half-Kelly is an improper score (optimal report $p^* = 2q - c$ exaggerates edge $2\times$, hitting Brief 2's wall); deploys $31–53\%$ of cash (cutting 52-wk power from 0.66 to 0.24); `cash + open cost ⩾ 20` is purchase-invariant and deadlocks if 1 settlement record is lost
3	**Critical**	Kalshi $\$2,500$ stake & 7-day universe (§12, §5.3–5.4)	$\$2,500$ requires $5,000–25,000$ contracts; collapsing ladders on Kalshi's 7-day non-sports universe leaves $< 5–15$ events/wk (failing the 30-event/wk abort wall) and book-walk slippage wipes out 2–4 pt edges
4	**Critical**	Study 1 paired SPRT (`arithmetic.py.txt`, §6)	Enumeration hardcodes $p_{11} = p_A p_B$ (independence on the same market); across the allowed $[0.1, 0.3]$ calibration band and correlated markets, power drops to $0.486–0.685$ and Type-II error jumps to $14–31\%$
5	**High**	Study 2 validity & Property 2 (§5.1, §5.7, `account`)	Under proportional staking $s_t = f_t C_t$, delayed settlement recording shields payouts from drawdowns and inflates cash ($8.10 \rightarrow$

Rank	Severity	Target	Core Theoretical / Operational Break
			20.29); `account()` simulates simultaneous rather than sequential stakes
6	**High**	Study 1 alignment, pick & checks (§6, Finding 2)	Frozen pilot kernel's bankruptcy resurrection corrupts search-period return scores (the pick) and fills; scale-invariant cosine ignores fee ruin; Check 3 masks lookahead leaks up to transaction cost drag
7	**High**	Wall derivations & Table 1 Row G (§2.1–2.3, §3)	Compares 4.98 (80% power) with 26 ($\sim 50\%$ power, expected log wealth $= \ln 20$ under constant oracle bet); Row G puts $\sim 50\%$-power expected hitting times in an 80%-power table
8	**Medium**	`argon-paper-findings.md` (Findings 4, 7, 10, 11)	Wrongly rejects Finding 11 (funding persists in unconditional SPA/StepM null returns $r_t = \epsilon_t - f_t - c_t$); uses hidden $\frac{0.05}{99}$ Bonferroni in Finding 10; checks DSR ceiling only at $\hat{S} \to \infty, \hat{\gamma}_3 = 0$
9	**Medium**	`paper-changes.md` (A2.3, A5.5, A7.2, A7.3, B)	Propagates conflicting log-growth formulas (`e - 1/(2m)` vs. `e - 1/(2m) - e^2/2`), unstated 99-way Bonferroni cutoffs, and mirror-account fallacies into `whitepaper.tex`
10	**Medium**	Human's §12 answers (search API, Polymarket, perps)	Search APIs returning full third-party page text cannot grant storage/LLM licenses over publisher copyright; retaining scraped Polymarket data violates August 2026 ToS
